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Improving a chain theorem for triangle-free 3-connected matroids

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ABSTRACT

In 2007, Kriesell established a chain theorem for triangle-free 3-connected graphs. Any triangle-free 3-connected graph can be reduced to a double-wheel or to $K_{3,3}$ by performing a sequence of simple operations without leaving the class of triangle-free 3-connected graphs. Double-wheels define the only infinite family of graphs that are irreducible with respect to these simple operations. In 2013, Lemos extended Kriesell's theorem for matroids. In this case, there are four infinite families of irreducible matroids. In this paper, we improve these results by proving that one of Kriesell's reduction operations can be avoided provided the number of families of irreducible matroids is increased by four.

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1. Introduction

With rare exceptions, for matroid theory, we use the notation and terminology set by Oxley [14]. For a natural n , we use $[n]$ to denote the set $\{1, 2, \dots, n\}$. Tutte [17] established that wheels and whirls are the unique 3-connected matroids having only essential elements. Tutte's theorem can be restated as:

Theorem 1.1. *If M is a 3-connected matroid having at least 4 elements, then there is a sequence of non-empty 3-connected matroids $M_0, M_1, M_2, \dots, M_n$ such that M_0 is isomorphic to a wheel or whirl, $M_n = M$, and, for every i in $[n]$, M_i is a single-element extension or a single-element coextension of M_{i-1} .*

Therefore we can construct any 3-connected matroid starting from an appropriate wheel or whirl by performing a sequence of single-element extensions or coextensions without leaving the class of 3-connected matroids. This sentence can be rewritten in reverse order. It is possible to reduce any

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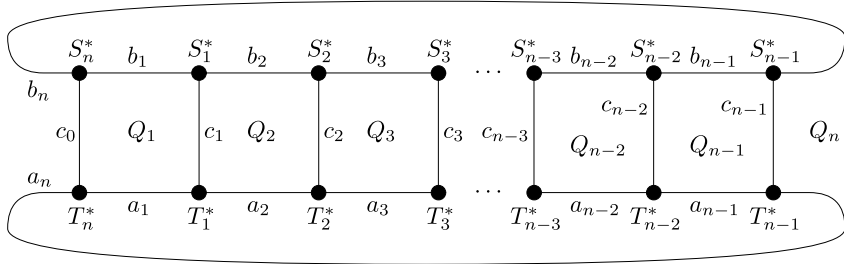


Fig. 1. The ladder L_n with $2n$ vertices.

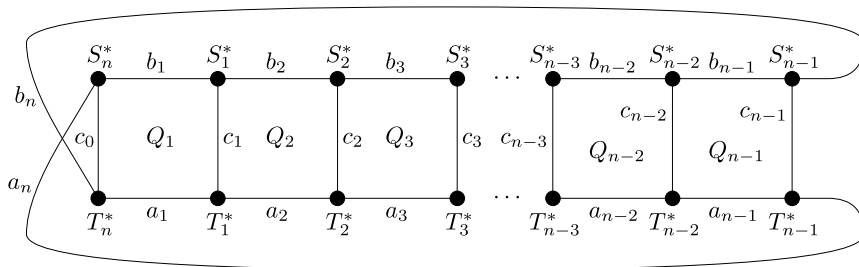


Fig. 2. The Möbius ladder L_n with $2n$ vertices.

3-connected matroid to a wheel or whirl by performing a sequence of single-element contractions or deletions without leaving the class of 3-connected matroids. Lemos [12] proved a similar result for the class of triangle-free 3-connected matroids that generalizes a result proved by Kriesell [11] for graphs. In this paper, we improve Lemos's theorem.

A circuit of a matroid with cardinality 4 is said to be a *square*. Squares and triads play a very important role in the proof of the main result of this paper. This role is similar to the one played by triangles and triads in the proof of Tutte's Wheels and Whirls Theorem.

First, we define a sequence of operations which can be performed in a triangle-free 3-connected matroid to reduce its size. Let M be a triangle-free 3-connected matroid having at least 5 elements. For each i in $\{1, 2, 3, 4\}$, we define when M is i -reducible.

We say that M is 1 -reducible or 2 -reducible provided M has an element e such that $M \setminus e$ or M/e respectively is a triangle-free 3-connected matroid.

We say that M is 3 -reducible when M has squares Q_1 and Q_2 such that $Q_1 \cap Q_2 = \{f\}$, for some element f belonging to a unique triad $T^* = \{e, f, g\}$ of M , and $M/e \setminus f$ is a triangle-free 3-connected matroid. We say that M is 4 -reducible when M has squares Q_1 and Q_2 such that $Q_1 \cap Q_2 = \{f\}$, for some element f belonging to exactly two triads $T^* = \{e, f, g\}$ and $T'^* = \{e', f, g'\}$ of M and $M/\{e, e'\} \setminus f$ is a triangle-free 3-connected matroid. (The last two reductions are more restrictive than the ones used by Kriesell [11]. In his definitions, the existence of the squares Q_1 and Q_2 are not required.)

Let K be a matroid such that $|E(K)| \geq 5$. When I is a subset of $\{1, 2, 3, 4\}$, we say that K is I -irreducible provided K is a triangle-free 3-connected matroid that is not i -reducible for every $i \in I$. To avoid a cumbersome notation, we use $abc \dots z$ -irreducible instead of $\{a, b, c, \dots, z\}$ -irreducible. Now, we define the families of matroids appearing in Theorem 1.2.

The graph illustrated in Fig. 1 is the ladder L_n with $2n$ vertices, for $n \geq 4$.

We use the labeling indicated in Figs. 1 and 2 in Section 2. The graph illustrated in Fig. 2 is the Möbius ladder L_n with $2n$ vertices, for $n \geq 4$.

For $n \geq 4$, let A_n be the auxiliary graph displayed in Fig. 3. Set $D = \{a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_n\}$. The relaxed non-binary ladder R_n of rank $2n$ is a matroid over $E(A_n)$ such that $\mathcal{C}(R_n) = \mathcal{C} \cup \mathcal{D}$, where

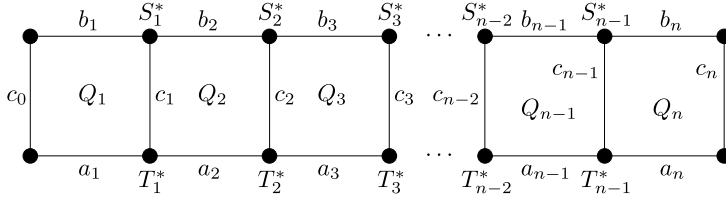


Fig. 3. An auxiliary graph A_n with $2n + 2$ vertices.

- (i) $C \in \mathcal{C}$ if and only if C is a circuit of the graphic matroid associated with A_n and $C \neq D \cup \{c_0, c_n\}$; and
- (ii) $C \in \mathcal{D}$ if and only if $C = E(T)$, where T is a tree of A_n such that: each leaf vertex of T is incident in A_n with c_0 or c_n ; and every vertex incident with c_0 or c_n in A_n is a vertex of T .

(It is implicit in the proof of Proposition 2.6 that R_n is a matroid.) Note that D is a basis of R_n such that, for every $i \in \{0, 1, 2, \dots, n\}$, $D \cup c_i$ is a circuit of R_n . There is a matroid P_n over $E(A_n)$ such that

$$C(P_n) = [C(R_n) - \{D \cup c_i : 0 \leq i \leq n\}] \cup \{D\}.$$

We say that P_n is the *non-binary ladder of rank $2n$* . Observe that D is a circuit-hyperplane of P_n and R_n is obtained from P_n by relaxing the circuit-hyperplane D . The main result of this paper is:

Theorem 1.2. *Let M be a 123-irreducible matroid with at least 11 elements. If M is 4-reducible, then M is isomorphic to $M(G)$, where G is a ladder or a Möbius ladder, or M is isomorphic to a non-binary ladder or M is isomorphic to a relaxed non-binary ladder.*

Therefore, in the main result of Lemos [12], we do not need to use the 4-reduction operation provided we increase the number of “irreducible” families of matroids. We derive the consequences of Theorem 1.2 and the main result of Lemos [12] in Section 5. In particular, in Section 5, we state a new chain theorem for the class of 3-connected triangle-free matroids.

Two families that appear in Theorem 1.2 also appear in the main result of Chun, Mayhew and Oxley [1]. The idea of our result’s proof is similar to the proof of their theorem but we do not use any of their intermediary lemmas because we are not dealing with binary matroids.

Results similar to Theorem 1.1 have been proved by Seymour [16], Coullard and Oxley [7], Geelen and Whittle [8], Hall [10], Geelen and Zhou [9], Oxley, Semple and Whittle [15], Mayhew, Royle and Whittle [13], Chun, Mayhew and Oxley [1–6], for example.

2. Constructing some matroids from their circuits and cocircuits

In Section 4, we obtain many triads and squares for a 4-reducible 123-irreducible triangle-free 3-connected matroid. In this section, we establish that these triads and squares are enough to determine the four families of matroids that appear in Theorem 1.2.

Proposition 2.1. *Let $a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_n, c_1, c_2, \dots, c_n$ be pairwise different elements of a 3-connected matroid M , for an integer n exceeding 3. If*

- (i) *for each $k \in [n]$, $Q_k = \{a_k, b_k, c_{k-1}, c_k\}$ is a square of M (take $c_0 = c_n$);*
- (ii) *for each $k \in [n - 1]$, $S_k^* = \{b_k, b_{k+1}, c_k\}$ and $T_k^* = \{a_k, a_{k+1}, c_k\}$ are triads of M ; and*
- (iii) *$S_n^* = \{x_n, b_1, c_n\}$ and $T_n^* = \{y_n, a_1, c_n\}$ are triads of M , where $\{x_n, y_n\} = \{a_n, b_n\}$,*

then $M/Y = M(L_n)$, when $x_n = b_n$ and $y_n = a_n$, or $M/Y = M(\mathcal{L}_n)$, when $x_n = a_n$ and $y_n = b_n$, where $Y = E(M) - \{a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_n, c_1, c_2, \dots, c_n\}$. Moreover, $|Y| \leq 1$ and, when $Y \neq \emptyset$, $Y \cup \{a_i, b_i\}$ is a triad of M , for every $i \in [n]$.

The graph illustrated in Fig. 1 is the ladder L_n with $2n$ vertices. When $x_n = b_n$ and $y_n = a_n$, the triads listed in items (ii) and (iii) of Proposition 2.1 are the vertex-triads of L_n . In this case, we also indicate in Fig. 1 the squares listed in (i) of Proposition 2.1.

The graph illustrated in Fig. 2 is the Möbius ladder $\mathcal{L}_n$ with $2n$ vertices. When $x_n = a_n$ and $y_n = b_n$, the triads listed in items (ii) and (iii) of Proposition 2.1 are the vertex-triads of $\mathcal{L}_n$. In this case, with the exception of Q_n , we also indicate in Fig. 2 the squares listed in (i) of Proposition 2.1. Note that Q_n is also a square of $\mathcal{L}_n$ but it is not a “face” in this drawing.

Proof. Set $X = \{a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_n, c_1, c_2, \dots, c_n\}$. For each $i \in [n]$ and $k \in [n-1]$, consider

$$C_{i,i+k} = \{a_{i+1}, a_{i+2}, \dots, a_{i+k}, b_{i+1}, b_{i+2}, \dots, b_{i+k}, c_i, c_{i+k}\},$$

where the indices are taken modulo n . Observe that $Q_{i+1} = C_{i,i+1}$ and, when $k \leq n-2$,

$$[C_{i,i+k} \cup Q_{i+k+1}] - c_{i+k} = C_{i,i+k+1}.$$

Therefore, by induction and the exchange axiom for circuits, $C_{i,i+k}$ is a dependent set of M . By orthogonality,

$$C_{i,i+k} \text{ is a circuit of } M. \quad (2.1)$$

Sublemma 2.2. If $C \in \mathcal{C}(M|X)$ and $\{a_j, b_j\} \subseteq C$, for some $j \in [n]$, then $C = C_{i,i+k}$, for some $i \in [n]$ and $k \in [n-1]$, or $\{a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_n\} \subseteq C$.

Proof. There are $i \in [n]$ and $k \in [n]$ such that

$$\{a_j, b_j\} \subseteq \{a_{i+1}, a_{i+2}, \dots, a_{i+k}, b_{i+1}, b_{i+2}, \dots, b_{i+k}\} \subseteq C.$$

Choose i and k such that k is maximum. If $k = n$, then the result follows. Assume $k < n$. By the choice of i and k , $|\{a_i, b_i\} \cap C| \leq 1$. By orthogonality with S_i^* and T_i^* , $c_i \in C$. Similarly, $c_{i+k} \in C$. Thus $C_{i,i+k} \subseteq C$. The result follows from (2.1). $\square$

Consider $\mathcal{C}_1 = \{C \in \mathcal{C}(M|X) : C \cap \{a_i, b_i\} = \emptyset, \text{ for some } i \in [n]\}$. First, we establish that

$$\mathcal{C}_1 = \{C_{i,i+k} : i \in [n] \text{ and } k \in [n-1]\}. \quad (2.2)$$

If $C \in \mathcal{C}_1$, then there is an $j \in [n]$ such that $C \cap \{a_j, b_j\} = \emptyset$. By symmetry, we may assume that $j = 1$. If $C \cap \{c_1, c_2, \dots, c_{i-1}\} = \emptyset$, then, by orthogonality, $C \cap \{a_1, a_2, \dots, a_i, b_1, b_2, \dots, b_i\} = \emptyset$. Therefore there is $i \in [n]$ such that $c_i \in C$. Choose i as small as possible. In particular, $C \cap \{a_1, a_2, \dots, a_i, b_1, b_2, \dots, b_i\} = \emptyset$. By orthogonality, $\{a_{i+1}, b_{i+1}\} \subseteq C$. By Sublemma 2.2, $C = C_{i,i,k}$, for some $k \in [n-1]$. Thus (2.2) follows.

Consider $\mathcal{C}_2 = \{C \in \mathcal{C}(M|X) : |C \cap \{a_i, b_i\}| = 1, \text{ for every } i \in [n]\}$. We have:

Sublemma 2.3. If $C \in \mathcal{C}_2$ and $i \in [n]$, then $C \Delta Q_i \in \mathcal{C}_2$.

Proof. By symmetry, we may assume $a_i \in C$ and $i = 2$. By the exchange axiom for circuits, there is a circuit D of M such that $b_2 \in D \subseteq (C \cup Q_2) - a_2$. As $|D \cap \{a_j, b_j\}| \leq 1$, for every $j \in [n]$, it follows, by (2.2), that $D \notin \mathcal{C}_1$. Hence $D \cap \{a_j, b_j\} \neq \emptyset$, for every $j \in [n]$, and so $D \in \mathcal{C}_2$. In particular, for $j \in [n] - \{2\}$, $\{a_j, b_j\} \cap C = \{a_j, b_j\} \cap D$. By orthogonality with S_1^* and T_1^* , $c_1 \in C$ iff $b_1 \in C$ iff $b_1 \in D$ iff $c_1 \notin D$. Similarly, $c_2 \in C$ iff $c_2 \notin D$. As, for $j \in [n] - \{1, 2\}$, $C \cap (S_j^* - c_j) = D \cap (S_j^* - c_j)$ and $C \cap (T_j^* - c_j) = D \cap (T_j^* - c_j)$ are sets whose cardinalities add to 2, it follows, by orthogonality, that $c_j \in C$ iff $c_j \in D$. Therefore $D = C \Delta Q_i$. $\square$

Consider $\mathbf{V} = \{(v_1, v_2, \dots, v_n) : v_i \in \{a_i, b_i\}, \text{ for every } i \in [n]\}$. For $v = (v_1, v_2, \dots, v_n) \in \mathbf{V}$, we define

$$C_v = \{v_1, v_2, \dots, v_n\} \cup \{c_i : i \in [n-1] \text{ and } |S_i^* \cap \{b_i, b_{i+1}\}| = 1 \text{ or } i = n \text{ and } |S_n^* \cap \{b_1, x_n\}| = 1\}.$$

Now, we prove that

$$\mathcal{C}_2 = \emptyset \text{ or } \mathcal{C}_2 = \{C_v : v \in \mathbf{V}\}. \quad (2.3)$$

Assume $\mathcal{C}_2 \neq \emptyset$. For $v = (v_1, v_2, \dots, v_n) \in \mathbf{V}$, choose $C \in \mathcal{C}_2$ such that $|C \cap \{v_1, v_2, \dots, v_n\}|$ is maximum. Our goal is to establish that $\{v_1, v_2, \dots, v_n\} \subseteq C$. If $v_i \notin C$, then, by [Sublemma 2.3](#), $C \Delta Q_i$ is a circuit of M contrary to the choice of C , since $(C \Delta Q_i) \cap \{v_1, v_2, \dots, v_n\} = (C \cap \{v_1, v_2, \dots, v_n\}) \cup v_i$. Therefore $\{v_1, v_2, \dots, v_n\} \subseteq C$. By orthogonality, $C = C_v$ and [\(2.3\)](#) follows.

Sublemma 2.4. If $\mathcal{C}_2 \neq \emptyset$, then,

- (i) $\{a_1, a_2, \dots, a_n\}$ and $\{b_1, b_2, \dots, b_n\}$ are circuits of M , when $x_n = b_n$ and $y_n = a_n$;
- (ii) for each $i \in [n]$, $\{a_1, \dots, a_i, c_i, b_{i+1}, \dots, b_n\}$ and $\{b_1, \dots, b_i, c_i, a_{i+1}, \dots, a_n\}$ are circuits of M , when $x_n = a_n$ and $y_n = b_n$; and
- (iii) $\{a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_n\}$ is a spanning circuit of M , when $x_n = a_n$ and $y_n = b_n$.

Proof. Note that (i) and (ii) are immediate consequences of [\(2.3\)](#). By (ii), [Sublemma 2.2](#) and the exchange axiom for circuits, we have (iii). $\square$

Sublemma 2.5. If $\mathcal{C}_2 \neq \emptyset$, then $M = M(L_n)$, when $x_n = b_n$ and $y_n = a_n$, or $M = M(\mathcal{L}_n)$, when $x_n = a_n$ and $y_n = b_n$.

Proof. By [\(2.2\)](#), [\(2.3\)](#) and [Sublemmas 2.2](#) and [2.4](#), $\mathcal{C}(M|X) = \mathcal{C}(M(L_n))$, when $x_n = b_n$ and $y_n = a_n$, or $\mathcal{C}(M|X) = \mathcal{C}(M(\mathcal{L}_n))$, when $x_n = a_n$ and $y_n = b_n$. The result follows provided we establish that $X = E(M)$. By [Sublemma 2.4](#), $\{a_1, a_2, \dots, a_{n-1}, b_1, b_2, \dots, b_{n-1}, c_n\}$ spans X in M . Therefore $r(X) \leq 2n - 1$. As $\{a_1, a_2, \dots, a_n, b_1\}$ spans X in M^* , it follows that $r^*(X) \leq n + 1$. Thus

$$r(X) + r^*(X) - |X| \leq (2n - 1) + (n + 1) - 3n = 0.$$

But M is 3-connected and so $X = E(M)$. $\square$

We may assume that

$$\mathcal{C}_2 = \emptyset \quad (2.4)$$

otherwise the result follows from [Sublemma 2.5](#).

Applying the elimination axiom for circuits to $\{a_2, a_3, \dots, a_n, b_2, b_3, \dots, b_n, c_1, c_n\}$ and Q_n , we conclude that $\{a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_n, c_1\}$ is a dependent set of M . Observe this set spans X in M and so $r(X) \leq 2n$. As $\{a_1, a_2, \dots, a_n, b_1\}$ spans X in M^* , it follows that $r^*(X) \leq n + 1$. Thus

$$r(X) + r^*(X) - |X| \leq (2n) + (n + 1) - 3n = 1.$$

Therefore $|E(M) - X| \leq 1$.

By [\(2.2\)](#), [\(2.4\)](#) and [Sublemma 2.2](#), $\{a_i, b_i\}$ is a series class of $M|X$, for every $i \in [n]$. Thus $E(M) - X \neq \emptyset$ because M is 3-connected. Therefore $|E(M) - X| = 1$, say $E(M) - X = \{e\}$. Moreover, for each $i \in [n]$, $R_i^* = \{a_i, b_i, e\}$ is a triad of M . If C is a circuit of M such that $e \in C$, then, by orthogonality, C meets $\{a_i, b_i\}$, for every $i \in [n]$. In particular, M/e is simple and Q_i is a circuit of M/e , for every $i \in [n]$. Now, we show that M/e is 3-connected. Assume that $\{V, W\}$ is a k -separation for M/e , for $k \leq 2$. As M/e is simple and cosimple, it follows that $\min\{|V|, |W|\} \geq 3$. Without loss of generality, we may assume that $|V \cap U^*| \geq 2$, for at least two $U^* \in \{S_1^*, S_2^*, T_1^*, T_2^*\}$. We may also assume that V is closed in M and M^* . Therefore V contains at least two triads in $\{S_1^*, S_2^*, T_1^*, T_2^*\}$. Hence $Q_2 \subseteq V$, because V is closed in M , and $S_1^* \cup S_2^* \cup T_1^* \cup T_2^* \subseteq V$, because V is closed in M^* . By induction, it is not difficult to show that $X \subseteq V$; a contradiction. Therefore M/e is 3-connected. By [Sublemma 2.5](#), $M/e = M(L_n)$, when $x_n = b_n$ and $y_n = a_n$, or $M/e = M(\mathcal{L}_n)$, when $x_n = a_n$ and $y_n = b_n$. The result follows. $\square$

From [Proposition 2.1](#), when $Y \neq \emptyset$, it is not difficult to construct M but we will not do it here because we do not need this information.

Proposition 2.6. Let $a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_n, c_0, c_1, c_2, \dots, c_n$ be pairwise different elements of a 3-connected matroid M , for an integer n exceeding 3. If

- (i) for each $k \in [n]$, $Q_k = \{a_k, b_k, c_{k-1}, c_k\}$ is a square of M ;
- (ii) for each $k \in [n-1]$, $S_k^* = \{b_k, b_{k+1}, c_k\}$ and $T_k^* = \{a_k, a_{k+1}, c_k\}$ are triads of M ; and
- (iii) $S^* = Q_1 - c_1$ and $T^* = Q_n - c_{n-1}$ are triads of M ,

then M/Y is isomorphic to a non-binary ladder or a relaxed non-binary ladder with rank- $2n$, where $Y = E(M) - \{a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_n, c_0, c_1, c_2, \dots, c_n\}$. Moreover, $|Y| \leq 1$.

Proof. Consider $X = \{a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_n, c_0, c_1, c_2, \dots, c_n\}$. Observe that $\{a_1, a_2, \dots, a_n, b_1\}$ spans X in M^* . If C^* is a cocircuit of M contained in $\{a_1, a_2, \dots, a_n, b_1\}$, then, by orthogonality with Q_i , $a_i \notin C^*$, for each $i \in \{2, 3, \dots, n\}$, and so $C^* \subseteq \{a_1, b_1\}$; a contradiction. Therefore $\{a_1, a_2, \dots, a_n, b_1\}$ is a coindependent set of M . Thus

$$r^*(X) = n + 1. \quad (2.5)$$

As $\{a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_n, c_0\}$ spans X in M , it follows that $r(X) \leq 2n + 1$. Hence there is an integer $\delta \geq 0$ such that

$$r(X) = 2n + 1 - \delta. \quad (2.6)$$

Thus

$$r(X) + r^*(X) - |X| = (2n + 1 - \delta) + (n + 1) - (3n + 1) = 1 - \delta \in \{0, 1\}.$$

As M is 3-connected, it follows that

$$|E(M) - X| = 1 - \delta \in \{0, 1\}. \quad (2.7)$$

For integers i and j such that $0 \leq i < j \leq n$, consider

$$C_{i,j} = \{a_{i+1}, a_{i+2}, \dots, a_{j-1}, a_j, b_{i+1}, b_{i+2}, \dots, b_{j-1}, b_j, c_i, c_j\}.$$

Similarly to the proof of Proposition 2.1, with the help of the exchange axiom for circuits and orthogonality, one can show that

$$C_{i,j} \text{ is a circuit of } M, \text{ when } i \neq 0 \text{ and } j \neq n. \quad (2.8)$$

Moreover,

$$C_{0,j} - c_0 \text{ is a circuit of } M, \text{ for } 0 < j < n, \text{ iff } C_{0,1} - c_0 \text{ is a circuit of } M. \quad (2.9)$$

But $C_{0,1} - c_0 = \{a_1, b_1, c_1\}$ cannot be a circuit of M because it is properly contained in Q_1 . Hence, using symmetry, when necessary, (2.9) implies

$$C_{0,j} - c_0 \text{ and } C_{j,n} - c_n \text{ are not a circuits of } M, \text{ for } 0 < j < n, \quad (2.10)$$

Thus (2.8) can be refined to

$$C_{i,j} \text{ is a circuit of } M, \text{ when } \{i, j\} \neq \{0, n\}, \quad (2.11)$$

since $C_{i,j}$ is a dependent set of M . Consider $D = \{a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_n\}$.

Sublemma 2.7. If $D \cup c_i$ is a circuit of M , for some i such that $0 \leq i \leq n$, then $D \cup c_j$ is a circuit of M , for every j such that $0 \leq j \leq n$. Moreover, $E(M) = X$ and there is a matroid N having D as a circuit-hyperplane such that M is obtained from N by relaxing D .

Proof. Assume this result fails. There is j , $0 \leq j \leq n$, such that $D \cup c_j$ is not a circuit of M . By symmetry, we may assume $i < j$. By the exchange axiom for circuits, there is circuit C of M such that $c_j \in C \subseteq [(D \cup c_i) \cup C_{i,j}] - c_i = D \cup c_j$. By the choice of j , there is k such that $a_k \notin C$ or $b_k \notin C$, say

$a_k \notin C$. By symmetry, we may assume $k < j$. By orthogonality, $\{a_1, a_2, \dots, a_j, b_1, b_2, \dots, b_j\} \cap C = \emptyset$. Again, by orthogonality, $C = C_{j,n} - c_n$; a contradiction to (2.10). Therefore $D \cup c_j$ is a circuit of M , for every j . By (2.6), D is a basis of M , as D spans X , and so $\delta = 1$. By (2.7), $E(M) = X$.

Consider the following family of subsets of X ,

$$\mathcal{D} = [C(M) - \{D \cup c_j : 0 \leq j \leq n\}] \cup \{D\}.$$

We will establish that $\mathcal{D}$ is the family of circuits of a matroid N over X . It enough to verify the exchange axiom for circuits for $\mathcal{D}$. Let W and Z be different elements of $\mathcal{D}$ such that $f \in W \cap Z$. Suppose that $W \cup Z - f$ does not contain an element of $\mathcal{D}$. If W and Z are circuits of M , then $W \cup Z - f$ contains a circuit Y of M . If $Y \in \mathcal{D}$, then $W \cup Z - f$ contains an element of $\mathcal{D}$; a contradiction. If $Y \notin \mathcal{D}$, then $D \subseteq Y$ and $W \cup Z - f$ contains an element of $\mathcal{D}$; a contradiction. Thus W or Z is not a circuit of M , say $W = D$. As $D \cup c_j$ is a circuit of M , for every j , it follows that $|Z - D| \geq 2$. Hence $|Z \cup D| \geq |D| + 2 = r(M) + 2$ and so $Z \cup D - f$ is a dependent set of M . Therefore $Z \cup D - f$ contains a circuit Y of M . Note that $Y \in \mathcal{D}$ because $f \notin Y$; a contradiction. Hence the matroid N exists. Note that D is a circuit-hyperplane of N . $\square$

Sublemma 2.8. *If $C_{0,n}$ is not a circuit of M , then $r(M) = 2n$ and D or $D \cup c_0$ is a circuit of M .*

Proof. By the exchange axiom for circuits, there are circuits of M whose union is equal to $C_{0,n} = (Q_1 \cup C_{1,n}) - c_1$. As $C_{0,n}$ is not a circuit of M , it follows that $C_{0,n} - c_n$ contains a circuit C of M . By orthogonality, $D \subseteq C$ and the result follows. $\square$

If D and $C_{0,n}$ are not circuits of M , then, by Sublemma 2.8, $D \cup c_0$ is a circuit of M . By Sublemma 2.7, there is a matroid N having D as a circuit-hyperplane such that M is obtained from N by relaxing D . Note that N satisfies the hypothesis of this proposition. Moreover, when the conclusion of this proposition holds for N , it also holds for M . Therefore, when necessary, we can replace M by N and we may assume that

$$D \text{ or } C_{0,n} \text{ is a circuit of } M. \quad (2.12)$$

Sublemma 2.9. *If C is a circuit of $M \setminus X$ such that $C \cap \{a_k, b_k\} = \emptyset$, for some $k \in [n]$, then $C = C_{i,j}$, for some i and j such that $0 \leq i < j \leq n$ and $\{i, j\} \neq \{0, n\}$.*

Proof. Assume this result fails. Choose a counter-example C such that $|C|$ is as small as possible. By orthogonality, $C \not\subseteq \{c_0, c_1, c_2, \dots, c_n\}$. Thus there is an integer i such that $C \cap \{a_i, b_i\} \neq \emptyset$. By symmetry, we may assume $i > k$. Subject to this condition, choose i such that i is as small as possible. By the choice of i , $C \cap \{a_{i-1}, b_{i-1}\} = \emptyset$. By orthogonality, $c_{i-1} \in C$ and $\{a_i, b_i\} \subseteq C$. As $C \neq Q_i = C_{i-1,i}$, it follows that $c_i \notin C$. By the elimination axiom for circuit, there is a circuit C' of M such that

$$c_i \in C' \subseteq (C \cup Q_i) - c_{i-1} = (C - c_{i-1}) \cup c_i.$$

By orthogonality, $C' \cap \{a_i, b_i\} = \emptyset$ and so $|C'| < |C|$. By the choice of C , $C' = C_{i,j}$, for some j such that $i < j \leq n$. (Remember that $c_i \in C'$.) But $C' \Delta Q_i = C_{i-1,j}$ is a subset of C . We arrive at a contradiction, by (2.11), since $i > k \geq 1$. $\square$

For $Z \subseteq \{0, 1, 2, \dots, n\}$, $Z \neq \emptyset$, say $Z = \{i_1, i_2, \dots, i_k\}$, where $i_1 < i_2 < \dots < i_k$, we define

$$\begin{aligned} A(Z) &= \left\{ a_i : 1 \leq i \leq i_1 \text{ or } i_k < i \leq n \text{ or } i_{2j-1} < i \leq i_{2j}, \text{ for some } 1 \leq j \leq \frac{k}{2} \right\} \\ B(Z) &= \left\{ b_i : 1 \leq i \leq i_1 \text{ or } i_k < i \leq n \text{ or } i_{2j} < i \leq i_{2j+1}, \text{ for some } 1 \leq j \leq \frac{k-1}{2} \right\} \\ C(Z) &= \{c_{i_1}, c_{i_2}, \dots, c_{i_k}\}. \end{aligned}$$

We set $C_Z = A(Z) \cup B(Z) \cup C(Z)$ and, when $|Z| \geq 2$,

$$D_Z = C_Z \Delta (\{a_i : i_1 < i \leq i_k\} \cup \{b_i : i_1 < i \leq i_k\}).$$

Consider the following subset of $\mathcal{C}(M|X)$:

$$C_1 = \{C_{i,j} : 0 \leq i < j \leq n \text{ and } \{i, j\} \neq \{0, n\}\} \cup (\mathcal{C}(M) \cap \{D, C_{0,n}\}).$$

Sublemma 2.10. *If $C \in \mathcal{C}(M|X) - C_1$, then $C = C_Z$ or $C = D_Z$, for some $Z \subseteq [n]$ such that $|Z| \geq 2$.*

Proof. Assume that $C \cap \{c_0, c_1, c_2, \dots, c_n\} = \{c_{i_1}, c_{i_2}, \dots, c_{i_k}\}$, where $0 \leq i_1 < i_2 < \dots < i_k \leq n$. If $k \leq 1$, then, by orthogonality, $C = D$ or $C = D \cup c_{i_1}$; a contradiction. Thus $k \geq 2$. By [Sublemma 2.9](#),

$$\{a_i, b_i\} \neq \emptyset, \text{ for every } i \in [n]. \quad (2.13)$$

First, we show that

$$\{a_i : 1 \leq i \leq i_1\} \cup \{b_i : 1 \leq i \leq i_1\} \subseteq C. \quad (2.14)$$

Assume $i_1 \geq 1$ otherwise (2.14) holds. By (2.13), a_1 or b_1 belongs to C , say a_1 . By orthogonality with S^* , $b_1 \in C$. By orthogonality, (2.14) holds. By symmetry,

$$\{a_i : i_k < i \leq n\} \cup \{b_i : i_k < i \leq n\} \subseteq C. \quad (2.15)$$

As $C \neq C_{i_j, i_{j+1}}$, it follows that a_i or b_i does not belong to C , for some i such that $i_j < i \leq i_{j+1}$. By orthogonality, $C \cap \{a_i : i_j < i \leq i_{j+1}\} = \emptyset$ or $C \cap \{b_i : i_j < i \leq i_{j+1}\} = \emptyset$. By (2.13), when $1 \leq j < k$,

$$C \cap (\{a_i : i_j < i \leq i_{j+1}\} \cup \{b_i : i_j < i \leq i_{j+1}\}) \subseteq \{a_i : i_j < i \leq i_{j+1}\} \cup \{b_i : i_j < i \leq i_{j+1}\}. \quad (2.16)$$

By (2.16) and orthogonality with $T_{i_j}^*$ and $S_{i_j}^*$, we have, for $1 < j < k$,

$$C \cap \{a_{i_j}, a_{i_j+1}, b_{i_j}, b_{i_j+1}\} \subseteq \{a_{i_j}, b_{i_j+1}\}, \{a_{i_j+1}, b_{i_j}\}. \quad (2.17)$$

The result is a consequence of (2.14), (2.15), (2.16) and (2.17). $\square$

Sublemma 2.11. *If $Z = \{i_1, i_2, \dots, i_k\}$, where $0 \leq i_1 < i_2 < \dots < i_k \leq n$, for $k \geq 2$, then*

- (i) C_Z is a circuit of M iff $C_{(Z-i_k) \cup i}$ is a circuit of M , where $i_{k-1} < i \leq n$, iff $C_{(Z-i_1) \cup j}$ is a circuit of M , where $0 \leq j < i_2$.
- (ii) D_Z is a circuit of M iff $D_{(Z-i_k) \cup i}$ is a circuit of M , where $i_{k-1} < i \leq n$, iff $D_{(Z-i_1) \cup j}$ is a circuit of M , where $0 \leq j < i_2$.
- (iii) C_Z is a circuit of M iff $C_{Z \cup i}$ is a circuit of M , where $i_k < i \leq n$, iff $D_{Z \cup j}$ is a circuit of M , where $0 \leq j < i_1$.
- (iv) D_Z is a circuit of M iff $D_{Z \cup i}$ is a circuit of M , where $i_k < i \leq n$, iff $C_{Z \cup j}$ is a circuit of M , where $0 \leq j < i_1$.

Proof. By symmetry, it is enough to establish (i) and (iii) for j when $j \neq i_1$. We start by showing (i). As the “only if” case implies the “if” case, it is necessary to prove the “only if” case only. Assume C_Z is a circuit of M . By (2.11) and the exchange axiom for circuits, there is a circuit C of M such that $c_j \in C \subseteq (C_Z \cup C_{j, i_1}) - c_{i_1}$. (When $j > i_1$, we define C_{j, i_1} to be $C_{i_1, j}$.) If $i_1 < j$, then $(C_Z \cup C_{j, i_1}) - c_{i_1} = C_{(Z-i_1) \cup j}$. By [Sublemma 2.10](#), $C = C_{(Z-i_1) \cup j}$ and the “only if” of (i) follows in this case. If $i_1 > j$, then $(C_Z \cup C_{j, i_1}) - c_{i_1} = C_{(Z-i_1) \cup j} \cup \{b_l : j < l \leq i_1\}$. By [Sublemma 2.10](#), $C = C_{(Z-i_1) \cup j}$ and the “only if” of (i) also follows in this case.

Next, we establish (iii). Assume C_Z is a circuit of M . By (2.11) and the exchange axiom for circuits, there is a circuit C' of M such that $c_j \in C' \subseteq (C_Z \cup C_{j, i_1}) - a_{i_1}$. As $(C_Z \cup C_{j, i_1}) - a_{i_1} = D_{Z \cup j} \cup \{a_l : j < l < i_1\}$, it follows, by [Sublemma 2.10](#), that $C' = D_{Z \cup j}$. Assume $D_{Z \cup j}$ is a circuit of M . By (2.11) and the exchange axiom for circuits, there is a circuit C'' of M such that $C'' \subseteq (D_{Z \cup j} \cup C_{j, i_1}) - c_j$. As $(D_{Z \cup j} \cup C_{j, i_1}) - c_j = C_Z$, it follows, by [Sublemma 2.10](#), that $C'' = C_Z$. This concludes the proof of (iii). $\square$

Sublemma 2.12. *If D is a circuit of M , then $E(M) = X$ and*

$$\begin{aligned} \mathcal{C}(M) = \{D\} \cup \{C_{i,j} : 0 \leq i \leq n \text{ and } \{i, j\} \neq \{0, n\}\} \\ \cup \{C_Z : Z \subseteq [n] \text{ and } |Z| \geq 2\} \cup \{D_Z : Z \subseteq [n] \text{ and } |Z| \geq 2\}. \end{aligned} \quad (2.18)$$

Proof. By [Sublemma 2.8](#), $r(M) = 2n$. By (2.6) and (2.7), $E(M) = X$. As $(D - a_1) \cup \{a_0, a_n\}$ is a dependent set of M , it follows, by [Sublemma 2.10](#), that $D_{\{0,n\}}$ is a circuit of M . Similarly $C_{\{0,n\}}$ is a circuit of M . This result follows provided we establish that C_Z and D_Z are circuits of M , for every $Z \subseteq [n]$ such that $|Z| \geq 2$, say C_Z . Assume $Z = \{i_1, i_2, \dots, i_k\}$, where $i_1 < i_2 < \dots < i_k$. Applying [Sublemma 2.11](#)(i) two times to $C_{\{0,n\}}$, we conclude that $C_{\{i_1, i_2\}}$ is a circuit of M . Now, to finish the proof, we conclude that C_Z is a circuit of M by applying $k - 2$ times [Sublemma 2.11](#)(iii) to $C_{\{i_1, i_2\}}$.

Note that a matroid over X whose family of circuits is given by (2.18) is a non-binary ladder. A matroid obtained by relaxing the circuit-hyperplane D of this non-binary ladder is a relaxed non-binary ladder.

By [Sublemma 2.12](#), when D is a circuit of M , the result follows. We may assume that D is not a circuit of M . By (2.12), $C_{0,n}$ is a circuit of M . By (2.6) and (2.7), $r(M) = 2n + 1$ and $|E(M) - X| = 1$, say $E(M) - X = \{e\}$. If M/e is 3-connected, then, by [Sublemma 2.12](#), the result follows, since M/e satisfies the hypothesis of [Proposition 2.6](#) and $r(M/e) = 2n$. Assume M/e is not 3-connected. Let $\{V, W\}$ be a 2-separation for M/e . First, we show that

$$\min\{|V|, |W|\} \geq 3. \quad (2.19)$$

If (2.19) fails, then $|V| = 2$ or $|W| = 2$, say $|V| = 2$. Hence $T = V \cup e$ is a triangle of M . By orthogonality, no element $f \in X$ such that $\{f\}$ is the intersection of two triads of M contained in X belongs to T . Thus $V = \{c_0, c_n\}$; a contradiction to orthogonality. Therefore (2.19) holds. Assume $|S^* \cap V| \geq 2$. By (2.19), it is possible to choose $\{V, W\}$ such that V is closed in both M and M^* . In particular, $S^* \subseteq V$ and $Q_1 \subseteq V$. By induction on i , it is simple to prove that $Q_1 \cup \dots \cup Q_i \subseteq V$, since

$$\begin{aligned} \{a_i\} &= T_i^* - (Q_1 \cup \dots \cup Q_{i-1}); \{b_i\} = S_i^* - (Q_1 \cup \dots \cup Q_{i-1}); \\ \{a_i, b_i, c_i\} &= Q_i - (Q_1 \cup \dots \cup Q_{i-1}). \end{aligned}$$

Hence $X \subseteq V$ and $W = \emptyset$. With this contradiction, we finish the proof of [Proposition 2.6](#). $\square$

3. Known lemmas

From Lemos [12], we need the following results.

Lemma 3.1. *Let M be a 1-irreducible matroid such that $|E(M)| \geq 7$. If Q_1 and Q_2 are different squares of M , then $|Q_1 \cap Q_2| \leq 2$. ([Lemma 2.2 of \[12\]](#))*

For a matroid M , we say that $F \subseteq E(M)$ is a *forced set* provided, for every $e \in E(M) - F$ and 2-separation $\{X, Y\}$ for any $N \in \{M \setminus e, M/e\}$, there is $Z \in \{X, Y\}$ such that $F \subseteq \text{fcl}_N(Z)$. For example, if, for every $W \subseteq F$, $F \subseteq \text{fcl}_M(W)$ or $F \subseteq \text{fcl}_M(F - W)$, then F is a forced set of M .

Lemma 3.2. *Suppose that F is a forced set of a triangle-free 3-connected matroid M . If $e \in E(M) - F$ and $e \in \text{cl}_{M^*}(F)$, then M/e is 3-connected. Moreover, when M is 2-irreducible, there is a square Q of M such that $e \in Q$. ([Lemma 4.2 of \[12\]](#)).*

Let M be a 3-connected matroid with at least four elements. We say that L^* is a *coline* of M provided L^* is a line of M^* . When $|L^*| \geq 3$, any 3-subset of L^* is a triad of M .

Lemma 3.3. *Suppose that F is a forced set of a triangle-free 3-connected matroid M . If $e \in E(M) - F$ and $e \in \text{cl}_M(F)$, then,*

- (i) every 2-separation for $M \setminus e$ is trivial and so $\text{co}(M \setminus e)$ is 3-connected; or
- (ii) e belongs to a coline of M with at least 4 elements. ([Lemma 4.3 of \[12\]](#)).

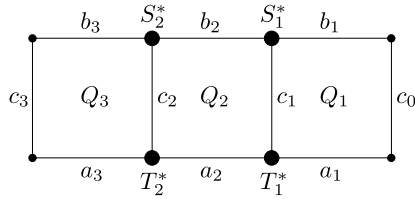


Fig. 4. Triads and squares of Proposition 4.1.

4. Proof of Theorem 1.2

Proposition 4.1. Suppose that M is a 123-irreducible matroid having at least 11 elements. If

- (i) there are squares Q_1 and Q_2 of M such that $|Q_1 \cap Q_2| = 1$, say $Q_1 \cap Q_2 = \{c_1\}$;
- (ii) c_1 belongs to exactly two triads of M , say $T_1^* = \{a_1, a_2, c_1\}$ and $S_1^* = \{b_1, b_2, c_1\}$, where a_1 and b_1 are different elements of Q_1 and a_2 and b_2 are different elements of Q_2 ; and
- (iii) $M \setminus c_1 / \{a_1, b_1\}$ is a triangle-free 3-connected matroid,

then $Q_1 - c_1$ or $Q_2 - c_1$ is not a triad of M and, when $Q_2 - c_1$ is not a triad of M ,

- (iv) there is a square Q_3 of M such that $Q_3 \cap (Q_1 \cup Q_2) = \{c_2\}$;
- (v) there are different elements of $Q_3 - c_2$, say a_3 and b_3 , such that $T_2^* = \{a_2, a_3, c_2\}$ and $S_2^* = \{b_2, b_3, c_2\}$ are the unique triads of M containing c_2 ; and
- (vi) $M \setminus c_2 / \{a_2, b_2\}$ is a triangle-free 3-connected matroid.

Proof. Let c_0 and c_2 be respectively the element belonging to $Q_1 - \{a_1, b_1, c_1\}$ and $Q_2 - \{a_2, b_2, c_1\}$. We illustrate these squares and triads in Fig. 4.

First, we show that

$$\text{at most one of } Q_1 - c_1 \text{ and } Q_2 - c_1 \text{ is a triad of } M. \quad (4.1)$$

If (4.1) fails, then $Q_1 \cup Q_2$ is spanned in M and M^* by respectively $\{a_1, a_2, b_1, b_2, c_1\}$ and $\{a_1, a_2, b_1\}$. Hence

$$r(Q_1 \cup Q_2) + r^*(Q_1 \cup Q_2) - |Q_1 \cup Q_2| \leq 5 + 3 - 7 = 1$$

and so $|E(M) - (Q_1 \cup Q_2)| \leq 1$; a contradiction. Thus (4.1) holds. Now, to prove the second part of this result, we assume that $Q_2 - c_1$ is not a triad of M .

If L^* is a coline of M such that $c_2 \in L^*$, then, by orthogonality, $|L^* - Q_2| \leq 1$. By (ii), $c_1 \notin L^*$ or $|L^*| = 2$. As $Q_2 - c_1$ is not a triad of M , it follows that $|L^* \cap Q_2| = |L^* \cap (Q_2 - c_1)| \leq 2$. Thus $|L^*| \leq 3$.

As $(Q_1 \cup Q_2) - c_2$ is a forced set, it follows, by Lemma 3.3, that

$$\text{co}(M \setminus c_2) \text{ is 3-connected.} \quad (4.2)$$

By (4.2), there is a triad T_2^* of M such that $c_2 \in T_2^*$, since M is 1-irreducible. By orthogonality with Q_2 , T_2^* contains a_2, b_2 or c_1 . By hypothesis, c_1 belongs to just two triads of M , T_1^* and S_1^* . Without loss of generality, we may assume that $a_2 \in T_2^*$. By orthogonality with Q_1 , $T_2^* \cap Q_1 = \emptyset$. Hence

$$T_2^* = \{a_2, a_3, c_2\}, \text{ for some } a_3 \notin Q_1 \cup Q_2, \quad (4.3)$$

since $Q_2 - c_1$ is not a triad of M . Similarly, there is a triad T_0^* of M such that $c_0 \in T_0^*$. It may happen that $T_0^* = Q_1 - c_1$.

By Lemma 3.2 applied to the forced set $Q_1 \cup Q_2$, there is a square Q_3 of M such that $a_3 \in Q_3$. Now, we show that

$$Q_3 \cap (Q_1 \cup Q_2) \in \{\{c_2\}, \{c_0, c_2\}, \{a_2, b_1, c_1\}\}. \quad (4.4)$$

If $c_1 \notin Q_3$, then, by orthogonality, $Q_3 \cap (S_1^* \cup T_1^*) = \emptyset$ because, by hypothesis, $M \setminus c_1 / \{a_1, b_1\}$ is a triangle-free 3-connected matroid. In this case, we have (4.4). Assume $c_1 \in Q_3$. By orthogonality with T_1^* and S_1^* , there are $e \in \{a_1, a_2\}$ and $f \in \{b_1, b_2\}$, such that $Q_3 = \{a_3, c_1, e, f\}$. By orthogonality with T_2^* , $e = a_2$. By Lemma 3.1, $|Q_2 \cap Q_3| \leq 2$. Hence $f = b_1$. Thus (4.4) follows. Next, we establish that:

Sublemma 4.2. *If $|Q \cap (Q_1 \cup Q_2)| \leq 2$, for every square Q of M such that $a_3 \in Q$, then c_2 belongs to at least two triads of M .*

Proof. Assume this result fails. Thus T_2^* is the unique triad of M containing c_2 . Therefore $\text{co}(M \setminus c_2) = M \setminus c_2 / a_3$. By (4.4), $Q_2 \cap Q_3 = \{c_2\}$, since, by hypothesis, $|Q_3 \cap (Q_1 \cup Q_2)| \leq 2$. By (4.2), $\text{co}(M \setminus c_2)$ is 3-connected. As M is 3-irreducible, it follows that $\text{co}(M \setminus c_2)$ contains a triangle T . Hence $T \cup a_3$ is a square of M . In the previous argument, we can take Q_3 to be $T \cup a_3$. By (4.4), $(T \cup a_3) \cap (Q_1 \cup Q_2) = \{a_2, b_1, c_1\}$ because $c_2 \notin T \cup a_3$; a contradiction to hypothesis. $\square$

Now, we refine (4.4) by showing that:

Sublemma 4.3. $Q_3 \cap (Q_1 \cup Q_2) \in \{\{c_2\}, \{c_0, c_2\}\}$.

Proof. Assume this result fails. By (4.4), $Q_3 = \{a_2, a_3, b_1, c_1\}$. Observe that $Q_1 \cup Q_2 \cup Q_3$ is spanned in M and M^* by respectively $\{a_1, a_2, b_1, b_2, c_1\}$ and $\{a_1, a_2, a_3, b_1, c_0\}$. Hence

$$r(Q_1 \cup Q_2 \cup Q_3) + r^*(Q_1 \cup Q_2 \cup Q_3) - |Q_1 \cup Q_2 \cup Q_3| \leq 5 + 5 - 8 = 2. \quad (4.5)$$

We must have equality in (4.5), since M is a 3-connected matroid having at least 11 elements. In particular, $\{a_1, a_2, a_3, b_1, c_0\}$ is a coindependent set of M . Therefore $T_0^* \neq Q_1 - c_1$ and $a_3 \notin T_0^*$. By orthogonality with Q_3 , $T_0^* \cap Q_1 = \{a_1, c_0\}$ and so T_0^* is the unique triad of M containing c_0 , say $T_0^* = \{a_0, a_1, c_0\}$. Similarly, there is a square Q_0 of M such that $a_0 \in Q_0$. If $Q_1 \cup Q_2 \cup Q_3$ spans a_0 in M , then $Q_1 \cup Q_2 \cup Q_3$ spans a_0 in both M and M^* and, by (4.5), $(Q_1 \cup Q_2 \cup Q_3) \cup a_0$ is a 2-separating set for M ; a contradiction because $|E(M)| \geq 11$. In particular, $|Q_0 \cap (Q_1 \cup Q_2)| \leq 2$. By symmetry we can apply Sublemma 4.2 to a_0 instead of a_3 , since Q_0 can be taken to be any square containing a_0 , to conclude that c_0 belongs to at least two triads of M ; a contradiction. $\square$

By Sublemma 4.3, the hypothesis of Sublemma 4.2 is satisfied because Q_3 can be chosen to be any square of M containing a_3 . By Sublemma 4.2, there is a triad S_2^* of M such that $c_2 \in S_2^*$ and $S_2^* \neq T_2^*$. Now, we prove that

$$S_2^* \text{ and } T_2^* \text{ are the unique triads of } M \text{ containing } c_2. \quad (4.6)$$

Let T^* be a triad of M containing c_2 . Remember that, by assumption, $Q_2 - c_1$ is not a triad of M . By orthogonality, $T^* \cap Q_2$ is equal to $\{a_2, c_2\}$ or $\{b_2, c_2\}$ or $\{c_1, c_2\}$. As S_1^* and T_1^* are the unique triads of M containing c_1 , it follows that $T^* \cap Q_2$ is equal to $\{a_2, c_2\}$ or $\{b_2, c_2\}$. But T_2^* and S_2^* are the unique triads of M meeting Q_2 in $\{a_2, c_2\}$ or $\{b_2, c_2\}$ respectively. We established (4.6). Let b_3 and c_3 be elements of M such that $S_2^* = \{b_2, b_3, c_2\}$ and $Q_3 = \{a_3, b_3, c_2, c_3\}$. Note that $b_3 \notin Q_1 \cup Q_2$ and, by Sublemma 4.3, if $c_3 \in Q_1 \cup Q_2$, then $c_3 = c_0$.

Sublemma 4.4. $\text{co}(M \setminus c_2) = M \setminus c_2 / \{a_2, b_2\}$ is a triangle-free

Proof. By (4.2), $\text{co}(M \setminus c_2)$ is 3-connected. By (4.6), $\text{co}(M \setminus c_2) = M \setminus c_2 / \{a_2, b_2\}$. To conclude this proof, we need to show that $M \setminus c_2 / \{a_2, b_2\}$ is triangle-free. Assume the contrary. Let T be a triangle of $M \setminus c_2 / \{a_2, b_2\}$. There is a circuit C of $M \setminus c_2$ such that $T = C - \{a_2, b_2\}$. By orthogonality, $|C \cap \{a_2, a_3\}| \neq 1$ and $|C \cap \{b_2, b_3\}| \neq 1$. As M is triangle-free, it follows that $a_3 \in T$ or $b_3 \in T$. By symmetry, we may assume that $a_3 \in T$. If $b_3 \notin T$, then C is a square of M containing a_3 and avoiding c_2 ; a contradiction to Sublemma 4.3. Hence $b_3 \in T$ and so $\{a_2, a_3, b_2, b_3\} \subseteq C$. By orthogonality with S_1^* and T_1^* , $C = \{a_2, a_3, b_2, b_3, c_1\}$. As $(Q_2 \cup Q_3) - c_3$ is a forced set, it follows, by Lemma 3.3, that there is a triad T^* of M such that $c_3 \in T^*$. By orthogonality with Q_3 , T^* contains a_3 or b_3 or c_2 . By (4.6), S_2^* and T_2^* are the unique triads of M containing c_2 . Hence T^* contains a_3 or b_3 . By orthogonality with C

and $Q_2, T^* = Q_3 - c_2$. Observe that $Q_2 \cup Q_3$ is spanned in M and M^* by respectively $\{a_2, a_3, b_2, b_3\}$ and $\{a_2, b_2, c_1, c_2\}$. Therefore

$$r(Q_2 \cup Q_3) + r^*(Q_2 \cup Q_3) - |Q_2 \cup Q_3| \leq 4 + 4 - 7 = 1;$$

a contradiction because M is 3-connected. $\square$

Sublemma 4.5. $c_0 \neq c_3$.

Proof. Suppose that $c_0 = c_3$. Observe that $\{b_1, b_2, b_3, c_0, c_1, c_2\}$ spans $Q_1 \cup Q_2 \cup Q_3$ in M . Hence $r(Q_1 \cup Q_2 \cup Q_3) \leq 6$. As M is 3-connected, it follows that

$$2 \leq r(Q_1 \cup Q_2 \cup Q_3) + r^*(Q_1 \cup Q_2 \cup Q_3) - |Q_1 \cup Q_2 \cup Q_3|$$

and so $r^*(Q_1 \cup Q_2 \cup Q_3) \geq 5$. Note that $\{a_1, a_2, a_3, b_1, c_0\}$ spans $Q_1 \cup Q_2 \cup Q_3$ in M^* . Thus $\{a_1, a_2, a_3, b_1, c_0\}$ is a coindependent set of M . In particular, $Q_1 - c_1$ is not a triad of M . By symmetry, similarly to (4.6), there are exactly two triads S_0^* and T_0^* of M containing c_0 . We may assume that $T_0^* = \{a_1, c_0, e\}$ and $S_1^* = \{b_1, c_0, f\}$. By orthogonality with Q_3 , $\{e, f\} = \{a_3, b_3\}$. We arrive at a contradiction because $\{a_1, a_2, a_3, b_1, c_0\}$ contains S_0^* or T_0^* . $\square$

Now, we finish the proof of Proposition 4.1. By Sublemmas 4.3 and 4.5, we have (iv). Observe that (v) follows from (4.6). Sublemma 4.4 implies (vi). $\square$

Proof of Theorem 1.2. By hypothesis, M is 4-reducible. Therefore there are squares Q_1 and Q_2 of M such that $|Q_1 \cap Q_2| = 1$, say $Q_1 \cap Q_2 = \{c_1\}$, c_1 belongs to exactly two triads of M , say S_1^* and T_1^* and $\text{co}(M \setminus c_1)$ is a triangle-free 3-connected matroid. It is possible to label the elements of Q_1 and Q_2 such that $Q_1 = \{a_1, b_1, c_0, c_1\}$, $Q_2 = \{a_2, b_2, c_1, c_2\}$, $S_1^* = \{b_1, b_2, c_1\}$ and $T_1^* = \{a_1, a_2, c_1\}$. (We use the same labeling as set in Proposition 4.1.) Let n be the largest integer such that $a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_n, c_0, c_1, c_2, \dots, c_n$ are pairwise different elements of M satisfying:

- (i) for every $i \in [n]$, $Q_i = \{a_i, b_i, c_{i-1}, c_i\}$ is a square of M ;
- (ii) for every $i \in [n-1]$, $S_i^* = \{b_i, b_{i+1}, c_i\}$ and $T_i^* = \{a_i, a_{i+1}, c_i\}$ are the unique triads of M containing c_i ; and
- (iii) for every $i \in [n-1]$, $\text{co}(M \setminus c_i)$ is a triangle-free 3-connected matroid.

By construction, $n \geq 2$. By Proposition 4.1, $n \geq 3$. For $n = 5$, we illustrate these squares and triads in Fig. 5.

Set $X_k = \{a_1, a_2, \dots, a_k, b_1, b_2, \dots, b_k, c_0, c_1, c_2, \dots, c_k\}$, where k is an integer such that $1 \leq k \leq n$. We divide the proof of this result into two cases.

Case 1. $Q_1 - c_1$ or $Q_n - c_{n-1}$ is not a triad of M .

By symmetry, we may assume that $Q_n - c_{n-1}$ is not a triad of M . By Proposition 4.1 applied to Q_{n-1} and Q_n , we conclude that:

- (iv) there is a square Q_{n+1} of M such that $Q_{n+1} \cap (Q_{n-1} \cup Q_n) = \{c_n\}$;
- (v) there are different elements of $Q_{n+1} - c_n$, say a_{n+1} and b_{n+1} , such that $T_n^* = \{a_n, a_{n+1}, c_n\}$ and $S_n^* = \{b_n, b_{n+1}, c_n\}$ are the unique triads of M containing c_n ; and
- (vi) $M \setminus c_n / \{a_n, b_n\}$ is a triangle-free 3-connected matroid,

Set $Q_{n+1} - \{a_n, b_n, c_n\} = \{c_{n+1}\}$. By the choice of n ,

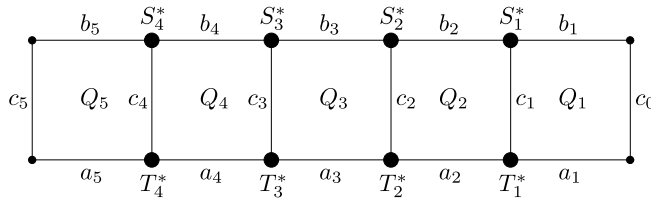
$$X_n \cap \{a_{n+1}, b_{n+1}, c_{n+1}\} \neq \emptyset. \quad (4.7)$$

By (iv), we can refine (4.7) to

$$X_{n-2} \cap \{a_{n+1}, b_{n+1}, c_{n+1}\} = X_n \cap \{a_{n+1}, b_{n+1}, c_{n+1}\} \neq \emptyset. \quad (4.8)$$

As $X_{n-2} \subseteq Q_1 \cup Q_2 \cup \dots \cup Q_{n-2} \subseteq X_{n-1}$, it follows, by orthogonality with S_{n+1}^* and T_{n+1}^* , that $a_{n+1}, b_{n+1} \notin X_{n-2}$, since $[(S_{n+1}^* - b_{n+1}) \cup (T_{n+1}^* - a_{n+1})] \cap X_{n-1} = \emptyset$. Thus (4.8) became

$$\{c_{n+1}\} = X_{n-2} \cap \{a_{n+1}, b_{n+1}, c_{n+1}\} = X_n \cap \{a_{n+1}, b_{n+1}, c_{n+1}\}. \quad (4.9)$$

Fig. 5. Triads and squares for $n = 5$.

As $X_{n-2} - c_0 \subseteq S_1^* \cup S_2^* \cup \dots \cup S_{n-2}^* \cup T_1^* \cup T_2^* \cup \dots \cup T_{n-2}^* \subseteq X_{n-1}$, it follows, by orthogonality and (4.9), that Q_{n+1} does not meet any triad in this union and so

$$c_0 = c_{n+1}. \quad (4.10)$$

By (4.9) and (4.10), $|X| = 3n + 3$, where

$$X = X_n \cup \{a_{n+1}, b_{n+1}\} = \{a_1, a_2, \dots, a_{n+1}, b_1, b_2, \dots, b_{n+1}, c_1, c_2, \dots, c_{n+1}\}.$$

By orthogonality with Q_1 , $Q_{n+1} - c_n$ is not a triad of M . By Proposition 4.1 applied to Q_n and Q_{n+1} , we conclude that $c_{n+1} = c_0$ belongs to exactly two triads, say S_{n+1}^* and T_{n+1}^* , where $S_{n+1}^* = \{b_{n+1}, c_0, x_1\}$ and $T_{n+1}^* = \{a_{n+1}, c_0, y_1\}$. By orthogonality with Q_1 , $\{x_1, y_1\} = \{a_1, b_1\}$, since S_1^* and T_1^* are the unique triads of M containing c_1 .

As M is 2-irreducible, it follows, by Proposition 2.1, that M is isomorphic to $M(L_{n+1})$ or $M(\mathcal{L}_{n+1})$. The result follows in this case.

Case 2. $Q_1 - c_1$ and $Q_n - c_{n-1}$ are triads of M .

In this case, the result follows from Proposition 2.6. $\square$

5. A chain theorem

To state the main result of Lemos [12], we need to define three more operations which can be performed in a triangle-free 3-connected matroid to reduce its size. Fix a triangle-free 3-connected matroid M having at least 5 elements.

We say that M is 5-reducible when M has squares Q_1 and Q_2 such that $Q_1 \cap Q_2 = \{f\}$, for some element f belonging to exactly three triads $T^* = \{e, f, g\}$, $T'^* = \{e', f, g'\}$ and $T''^* = \{e'', f, g''\}$ of M and N is a triangle-free 3-connected matroid, where N is obtained from $M/\{e, e', e''\} \setminus f$ after a $\Delta - Y$ operation along the triangle $\{g, g', g''\}$. This reduction is needed only for non-graphic matroids.

If there is a triad T^* and pairwise disjoint triads T_0^* , T_1^* and T_2^* of M such that $|T_i^* \cap T^*| = 1$, say $e_i \in T_i^* \cap T^*$, $T_i^* - e_i = \{f_i, g_i\}$ and $\{e_i, g_i, f_{i+1}, e_{i+1}\}$ is a square of M , for every $i \in \{0, 1, 2\}$, where the indices are taken modulo 3, and N is a triangle-free 3-connected matroid, where N is obtained from $M/\{g_0, g_1, g_2\} \setminus T^*$ after a $\Delta - Y$ operation along the triangle $\{f_0, f_1, f_2\}$, then we say that M is 6-reducible. In Fig. 6, we illustrate this reduction.

We say that M is 7-reducible when M has a triad T^* such that $M \setminus T^*$ is 3-connected and one of the following configurations is present around T^* :

- (i) There are pairwise different elements $e_0, e_1, e_2, f_0, f_1, f_2, g_0, g_1$ and g_2 of M such that $T^* = \{e_0, e_1, e_2\}$, $Q_i = \{e_i, g_i, f_{i+1}, e_{i+1}\}$ is a square of M and $Q_i - T^* \subseteq T_i^*$, for some triad T_i^* of M , for every $i \in \{0, 1, 2\}$, where the indices are taken modulo 3 (see the first diagram in Fig. 7).
- (ii) There are pairwise different elements e_0, e_1, e_2, f_0, f_1 and f_2 of M such that $T^* = \{e_0, e_1, e_2\}$, $Q_i = \{e_i, f_i, e_{i+1}, f_{i+1}\}$ is a square of M , for every $i \in \{0, 1\}$, and $T_1^* = \{f_1, f_2, f_3\}$ is a triad of M^* (see the second diagram in Fig. 7).

For example, in (ii), if $M = M(G)$, for some triangle-free 3-connected graph G , then T^* and T_1^* are the star of vertices with the same neighbors. Kriesell [11] does not ask the extra condition, namely (i) or (ii), on the triad T^* .

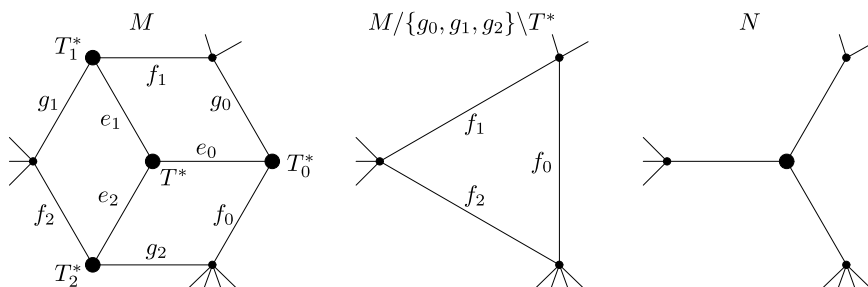
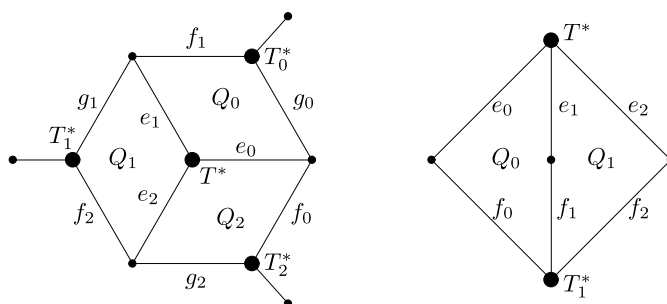


Fig. 6. The sixth reduction.

Fig. 7. The triad T^* to be deleted in the seventh reduction.

In Section 5 of [12], Lemos established the existence of exactly two non-isomorphic triangle-free 3-connected matroids whose ground set has $2n+1$ elements, for an integer $n \geq 6$, say $a_1, a_2, a_3, a_4, \dots, a_{2n}, a_{2n+1}$, having

- (i) $Q = \{a_1, a_2, a_3, a_4\}$ as a square; and
- (ii) for every $i \in \{1, 2, \dots, n\}$, $T_i^* = \{a_{2i-1}, a_{2i}, a_{2i+1}\}$ as a triad; and
- (iii) for every $i \in \{2, 3, \dots, n-1\}$, $Q_i = \{a_{2i-2}, a_{2i-1}, a_{2i+1}, a_{2i+2}\}$ as a square.

One of these matroids has $I = \{a_i : i \text{ is odd}\}$ as a circuit-hyperplane. It is called the *almost-double-wheel* with rank $n+1$. Relaxing I in it, we obtain the other matroid. It is called the *almost-double-whirl* with rank $n+1$.

In the next paragraph, we describe two families of matroids appearing in the main result of Lemos [12]. The first family contains only graphic matroids with even rank. The second family contains only non-graphic matroids with odd rank.

In Section 7 of Lemos [12], Lemos established the existence of a triangle-free 3-connected matroid whose ground set has $2n$ elements, for a integer $n \geq 5$, say $a_0, a_1, a_2, a_3, \dots, a_{2n-1}$, having $\{a_{2i-1}, a_{2i}, a_{2i+1}\}$ and $\{a_{2i-2}, a_{2i-1}, a_{2i+1}, a_{2i+2}\}$ as a triad and a square respectively, for every i , where the indexes are taken modulo $2n$. This matroid is binary. Now, we describe a way to obtain this matroid. Consider the labeling for the edges of a wheel W_n with n spokes given in the next picture. Note that $M(W_n)$ has the same triads and squares than the triangle-free matroid described in the beginning of this paragraph. But $M(W_n)$ has a lot of triangles. We destroy all these triangles by making a one-element coextension of $M(W_n)$. There is just one binary matroid N whose ground set is equal to $E(W_n) \cup e$, for a new element e , such that N is triangle-free and $N/e = M(W_n)$. In such matroid N , $\{a_1, a_3, a_5, a_7, \dots, a_{2n-1}\} \cup e$ must be a cocircuit and so N is unique in the class of binary matroids. Observe that $N \setminus e$ is a triangle-free matroid with all properties described in the beginning of this paragraph. If n is even, then there is a graph B_n such that $M(B_n) = N \setminus e$. This graph is the *double-wheel*

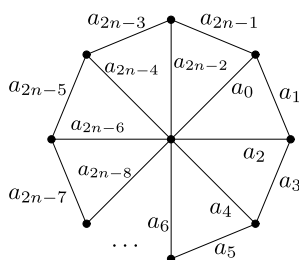


Fig. 8. A wheel with $2n$ edges.

defined by Kriesell [11]. When n is odd, Mayhew, Royle and Whittle [13] denoted N by Υ_n and called N as the *rank- n triadic Möbius matroid*. We say that e is the *tip* of N (see Fig. 8).

We can summarize Theorems 1.3 and 1.5 of Lemos [12] in the next result:

Theorem 5.1. *If M is an 1234567-irreducible matroid at with least 14 elements, then M is isomorphic*

- (i) *to a graphic matroid of a double-wheel; or*
- (ii) *to a matroid obtained from a triadic Möbius matroid deleting its tip; or*
- (iii) *to an almost-double-wheel; or*
- (iv) *to an almost-double-whirl.*

The next result is an immediate consequence of Theorems 1.2 and 5.1:

Theorem 5.2. *If M is an 123567-irreducible matroid at with least 14 elements, then M is isomorphic*

- (i) *to the graphic matroid of a double-wheel; or*
- (ii) *to the graphic matroid of a ladder; or*
- (iii) *to the graphic matroid of a Möbius ladder; or*
- (iv) *to the matroid obtained from a triadic Möbius matroid deleting its tip; or*
- (v) *to an almost-double-wheel; or*
- (vi) *to an almost-double-whirl; or*
- (vii) *to a non-binary ladder; or*
- (viii) *to a relaxed non-binary ladder.*

For the class of graphic matroids, we obtain the following result as a consequence of Theorem 5.2 because: it is impossible to perform the 5-reduction in a matroid belonging to this class; and the matroids described in items (iv) to (viii) are non-graphic.

Theorem 5.3. *If M is an 12367-irreducible graphic matroid at with least 14 elements, then M is isomorphic*

- (i) *to the graphic matroid of a double-wheel; or*
- (ii) *to the graphic matroid of a ladder; or*
- (iii) *to the graphic matroid of a Möbius ladder.*

This is a theorem about graphs because a 3-connected simple graph can be recovered, up to vertex labeling, from its associated matroid. Therefore Theorem 5.3 generalizes the main result of Kriesell [11], namely:

Theorem 5.4. *If M is an 123467-irreducible graphic matroid, then M is isomorphic*

- (i) *to the graphic matroid of a double-wheel; or*
- (ii) *to $M(K_{3,3})$.*

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